

Best practice guidance and reporting items for the development of scoping review protocols

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ABSTRACT

Objective: The purpose of this article is to clearly describe how to develop a robust and detailed scoping review protocol, which is the first stage of the scoping review process. This paper provides detailed guidance and a checklist for prospective authors to ensure that their protocols adequately inform both the conduct of the ensuing review and their readership.

Introduction: Scoping reviews are a common approach to evidence synthesis for researchers, clinicians, and policymakers across a variety of fields. Scoping reviews are not concerned with making analytical comparisons based on pooling results data from multiple primary sources of evidence, but rather on collating and describing the evidence and presenting the summation in a clearly illustrated format. Methods for undertaking and reporting scoping reviews continue to be refined. Some prospective reviewers may be uncertain how to plan, structure, and report scoping review protocols, as there is little or no specific guidance for scoping review protocols yet available.

Methods: This guidance was developed by members of the JBI Scoping Review Methodology Group based on previous experience and expertise in developing scoping review and evidence synthesis methodologies, protocols, and reviews, as well as through experiences working with and guiding authors to develop scoping review protocols. Elements of a comprehensive scoping review protocol are outlined and explained in detail.

Conclusion: Knowledge users of evidence syntheses rely on clear and transparent reporting to understand and use the results of published work to drive evidence-based improvements within health care and beyond. It is hoped that readers will be able to use this guidance when developing protocols to assist them in planning future scoping reviews and to carry them out with a high degree of transparency.

Keywords: evidence synthesis; evidence-based health care; PRISMA; protocol; scoping review

JBI Evid Synth 2022; 20(4):953–968.

Introduction

Scoping reviews are a fast-growing evidence synthesis approach for investigating and providing an overview of evidence and evidence gaps across a

variety of fields. The first formal framework for scoping reviews (then and sometimes still called “scoping studies”) was published in 2004.¹ In 2014, an international, multidisciplinary group of clinicians and evidence synthesis experts developed new guidance for the conduct and reporting of scoping reviews.^{2,3} This guidance was most recently updated in 2020,⁴ and now clearly reflects a relationship with the recent and complementary

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The authors declare no conflicts of interest.

DOI: 10.11124/JBIES-21-00242

Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR).⁵

As with other JBI evidence synthesis methodologies,⁶ the development of a protocol is essential prior to the conduct of a scoping review that uses JBI guidance. As Tricco and colleagues⁷ found in their 2016 scoping review, few scoping reviews (13%) reported using an a priori protocol. This may be because, prior to JBI's guidance, earlier guidance for scoping reviews did not specify that a protocol could or should be developed.^{1,8} Earlier studies that have examined the nature of scoping reviews have also not addressed the issue of protocols.^{9,10} While Pham and colleagues¹¹ followed an unpublished a priori protocol in their 2014 scoping review of scoping reviews and highlighted this as a strength, their review did not examine whether included scoping reviews used protocols themselves. Recognition of the value of scoping review protocols (and protocols in general) appears to be growing; the PRISMA-ScR encourages authors to report whether a protocol was developed but does not mandate that one must be developed. In this paper we argue that, as with systematic reviews, there are benefits to developing protocols for scoping reviews. We also provide guidance for authors to develop their own protocols for scoping reviews beyond the template presented in the *JBI Manual for Evidence Synthesis*.¹²

Protocols are widely regarded to be an important component to increasing value and reducing waste in research.^{13,14} In 2015, Shamseer and colleagues¹⁴ published the "Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015: elaboration and explanation," which referred to Moher and colleagues'¹⁵ 2007 work that noted that although several organizations required and published protocols, 89% of systematic reviews did not mention the development of a protocol. A priori review protocols are important because they provide a roadmap for reviewers regarding the direction and conduct of the resulting review. A protocol can be thought of as a recipe for a review that prespecifies the ingredients (inclusion criteria), where and how to find them (the databases and search strategy), and method. They can help reviewers avoid bias by reducing the risk of arbitrary decisions and unintended duplication of existing reviews, foster collaboration, and facilitate detection of selective reporting.¹⁴ Further, the registration or

publication of peer-reviewed protocols facilitates assessment and feedback from the research community.¹⁶ Reporting guidelines help authors to develop better manuscripts;¹⁶ therefore, as noted above, our intention is to support authors to write high-quality scoping review protocols to enable more transparently conducted and reported scoping reviews.

Scoping reviews are widely accepted by scientific and knowledge-user communities (people who are able to use the results of research for informed decisions about health policies, programs, and/or practices)¹⁷ as a legitimate and rigorous methodology for providing an overview on the state of evidence on subjects that are emerging, poorly known, interdisciplinary, complex, or dispersed across varied methodologies.⁴ Typically, the goal of a scoping review is to address broader questions compared with a traditional systematic review.¹⁸ As such, the steps common to scoping reviews do not correspond exactly to those of a systematic review. Notably, critical appraisal (also called assessment of risk of bias or methodological quality assessment) of the sources of evidence is often omitted because the main purpose of a scoping review is to describe and map a body of literature in terms of characteristics and factors detailed by the review's objective, question/s, and inclusion criteria.

Scoping reviews typically pose different kinds of questions than systematic reviews and thus have inclusion/exclusion criteria based on different parameters. For instance, because scoping reviews are typically more concerned with identifying broader, more descriptive elements of the literature rather than answering specific analytical questions of effectiveness, the inclusion criteria of "concept" and "context" may be used instead of "intervention" and "comparator." This is because scoping reviews do not aim to compare interventions in terms of pooled results effectiveness, but to descriptively map and explain the nature of a body of evidence. While scoping reviews can have quite specific questions and clearly defined inclusion/exclusion criteria, perhaps because of their descriptive rather than analytical objectives, scoping review searches often result in the identification of many potentially relevant sources of evidence.¹⁸ The task of screening and selecting sources to include from search results can be time consuming and unreliable without clearly articulated criteria. The development of a protocol that clearly describes the parameters of the review and the

decisions regarding inclusion and exclusion of evidence is therefore a valuable step in the conduct of scoping reviews.¹⁹

This paper focuses on the crucial preparatory stage of developing an a priori protocol for scoping reviews as a way of enhancing transparency and, thus, the utility and trustworthiness of the resulting review. It also provides prospective scoping review authors with more detailed guidance for developing protocols. With the ongoing proliferation of different types of evidence syntheses and notable variations in reporting standards and guidance across both systematic reviews and scoping reviews, clarity around the important features of scoping review protocols is necessary and timely.

Methods

This guidance was developed by members of the JBI Scoping Review Methodology Group based on previous experience and expertise in developing scoping review and evidence synthesis methodologies, protocols, and reviews, as well as through experiences working with and guiding authors to develop scoping review protocols.

Scoping review protocols

Scoping review protocols are unique documents that should be finalized prior to the conduct of the ensuing review. In this paper, we describe how protocols for scoping reviews can be developed based on adapted items of PRISMA-P.¹⁴ Table 1 maps the recommended items for a scoping review protocol onto the PRISMA-P and explains where specific adjustments may be made for scoping review protocols.^{20,21}

Depending on whether the protocol is submitted for publication in a journal or hosted elsewhere, such as Open Science Framework, Figshare, or Research-Gate, the presence and presentation of specific sections may differ based on local requirements.

All scoping review protocols should provide a comprehensive plan for the conduct and reporting of the subsequent scoping review report. The protocol is a valuable manuscript to publish or make publicly available because it informs the research community of the intentions and focus of the review authors prior to the conduct of the review itself. Each element outlined should ideally be provided in any scoping review protocol. These elements are

discussed in greater detail and important aspects to consider are explored here.

Title

As with any manuscript, the title is usually the first place readers will look to determine whether the content is relevant to their interests. Potential readers will also commonly use search functions of online databases to look for key words within manuscript titles. For these reasons, it is vital that authors include the most important words that describe the content of their protocol within the title itself. Words should be selected that most accurately describe the topic under review and align with the P (population), C (concept), C (context) framework (ie, the key words used in the inclusion criteria are usually key words within a descriptive title). Consideration should also be given to ensuring that the chosen words are those that are most used by (and therefore recognizable to) prospective readers. There may also be conventions around terminology with the topic area under review that may guide word choice (eg, “older people,” “older adults,” “elderly,” “geriatric,” “resident”). The words “scoping review” and “protocol” should also be included in the title to ensure that readers are able to easily determine the type of manuscript. An example of a good scoping review protocol title is: “Characteristics and validation of pediatric cancer screening and assessment tools: a scoping review protocol.”

Objective

A common issue in scoping review protocols is that the objective of the review is either unclear or lacking in specific detail. In line with the differences between traditional systematic reviews and scoping reviews described previously, the objective of a scoping review is also distinctive. The broad objective of all scoping reviews is to investigate and describe particular concepts or topics. This is compared with quantitative systematic reviews, which may pool numerical results to determine effectiveness, or qualitative systematic reviews, which examine and integrate participant experiences. Scoping review investigations may include extracting and charting information (eg, use of a particular concept) in relation to some other factor, or indeed a number of factors (eg, scientific field discipline or field, geographic region). While a systematic review with the objective of identifying the most accurate

Table 1: Recommended items to report in a scoping review protocol mapped against items in the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols Checklist²¹

Systematic review protocol			Scoping review protocol	
Section and topic	Item No	Checklist item	Information to report	Notes
ADMINISTRATIVE INFORMATION				
Title:				
Identification	1a	Identify the report as a protocol of a systematic review	Identify the report as a protocol of a scoping review	
Update	1b	If the protocol is an update of a previous systematic review, identify it as such	If the protocol is an update of a previous scoping review, identify it as such	
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and the registration number	If registered, provide the name of the registry (such as JBI) and the registration number	PROSPERO does not currently accept scoping review protocols for registration. We recommend Open Science Framework, Figshare, or Research-Gate as alternatives
Authors:				
Contact	3a	Provide name, institutional affiliation, email address of all protocol authors; provide physical mailing address of corresponding author		
Contributions	3b	Describe the contributions of the protocol authors and identify the guarantor of the review		
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		
Support:				
Sources	5a	Indicate sources of financial or other support for the review		
Sponsor	5b	Provide the name of the review funder and/or sponsor		
Role of sponsor or funder	5c	Describe the roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol		
INTRODUCTION				
Rationale	6	Describe the rationale for the review in the context of what is already known		Consider providing a rationale for the choice of conducting a scoping review as compared to other evidence synthesis approaches

Table 1: (Continued)

Systematic review protocol			Scoping review protocol	
Section and topic	Item No	Checklist item	Information to report	Notes
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Provide an explicit statement of the question(s) the review will address with reference to the inclusion/ exclusion criteria	Scoping reviews may not use PICO and instead may use JBI's Population, Concept, and Context (PCC) or another approach to reporting eligibility criteria
METHODS				
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, timeframe) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review		Scoping reviews may not use PICO and instead may use JBI's Population, Concept, and Context (PCC) or another approach to reporting eligibility criteria
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers, or other gray literature sources) with planned dates of coverage		
Search strategy	10	Present a draft of the search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated		
Study records:				
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review		Meta-analyses are not typically conducted in scoping reviews, so this can refer to the presentation of the results of the scoping review
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility, and inclusion in meta-analysis)	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility, and inclusion)	
Data collection process	11c	Describe the planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators		
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any preplanned data assumptions and simplifications		Scoping reviews may not use PICO and instead may use JBI's Population, Concept, and Context (PCC) or another approach to reporting eligibility criteria
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale		Scoping reviews may not extract outcome data, so this can refer to whichever data items are extracted

Table 1: (Continued)

Systematic review protocol			Scoping review protocol	
Section and topic	Item No	Checklist item	Information to report	Notes
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	If this is to occur, describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Scoping reviews typically do not include risk of bias assessment, but this information should be described if it will occur
Data synthesis	15a	Describe the criteria under which study data will be quantitatively synthesized	Describe criteria under which study data will be presented	Scoping reviews do not typically include quantitative synthesis of study data, but should still describe in advance how extracted data are anticipated to be presented in the resulting review
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (such as I ² , Kendall's τ)	Describe the planned approach to how extracted data will be presented (such as figures, tables, evidence gaps maps)	
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	Describe any proposed additional analyses (such as thematic analyses)	The JBI methodological guidance does not recommend undertaking thematic analysis, as this synthesis of data should ideally occur following methodological appraisal of the included sources
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned		
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)		Scoping reviews typically do not include assessment of meta-bias(es), but this information should be described if it will occur
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	If this is to occur, the method should be described	GRADE for scoping reviews currently does not exist, and at this stage it is unclear if a variation on GRADE would be useful

GRADE, Grading of Recommendations, Assessment, Development and Evaluation.

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, et al.; PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015;349:g7647. CC BY 4.0.

diagnostic test for a particular illness would be concerned with collation and comparison of the results of the test under study with those of the gold standard or reference test, a scoping review may instead be conducted with the objective of identifying all the screening and assessment tests used for a particular illness. In this case, the scoping review would focus on extracting the elements and questions from each test to establish commonalities and differences across the included tests in terms of the domains they measure, populations among which they have been used, and whether and in what ways they have been validated. Here it is clear that scoping reviews are not concerned with making analytical comparisons based on pooling results data from multiple primary sources of evidence, but rather on collating and describing the evidence and presenting the summation in a clearly illustrated format.

The objective needs to be clearly presented. An example of an appropriate objective for a scoping review could be:

In order to provide a comprehensive picture of pediatric cancer screening and assessment tools, this scoping review aims to identify and present the available information regarding screening and assessment tools used during the treatment of pediatric patients with cancer in terms of the issues/ domains measured or assessed, presence of validation, and how the tools have been validated.

From this objective, the scoping review is being conducted to describe the screening and assessment of pediatric cancer patients with respect to the instrument's validation, rather than being conducted to determine which tool is more accurate or which tool has the best psychometric validity. It also gives the reader a clear picture of what kinds of details are going to be extracted from included sources of evidence and how they will be reported in the resulting review. An inappropriate articulation of the same objective might be:

This scoping review aims to map the available evidence on pediatric cancer screening and assessment tools.

While some of the salient elements appear, the reader is left wondering what details will be extracted and why. Additionally, it would be almost impossible for a reader to assess the extent to which the objective of the review has been met. In the first objective above,

the reader has enough information to determine whether extracting and presenting these details does constitute the provision of “a comprehensive picture.” Having this level of detail is also extremely valuable to journal peer reviewers and editors because it provides an opportunity at the protocol stage to provide feedback regarding whether additional considerations should be included to truly meet the objective. For example, the authors may be advised to also include extraction and mapping of key details of the target populations addressed by the tools to provide a useful, comprehensive picture of screening and assessment tools in this area.

Review questions

The specific question or questions posed by the scoping review should be articulated in the protocol. These should be clearly linked to the objective of the review, and by answering the review's questions, the extent to which a review has met its objective may be assessed. The inclusion of specific questions allows the authors to go beyond the general objective statements and provide more precise details regarding the focus of the review and what will be examined and mapped. The PCC framework is considered appropriate when developing a scoping review research question. Some example questions that could be posed by the review with the objective presented earlier could be:

- i. What screening and assessment tools have been used to measure quality of life (concept) amongst pediatric patients with cancer (population) in tertiary institutions (context)?
- ii. What issues and/or domains do tools included for Question 1 address?
- iii. What specific population and cancer characteristics have been targeted by the tools (eg, population, tumor type, ages, sex, treatment type)?
- iv. Have the tools been validated and how?

Further detail could potentially be included for each of these questions if the authors wished, but with the detail already present, the reader is able to get a clear idea of the sorts of information that the review will be presenting and examining. This level of detail also allows the reader to evaluate the extent to which the objective may be met. For example, if the review provides answers to Questions 1–3, but does not provide clear details regarding the validation of the included tools, then there may be cause to

question whether the objective of providing a comprehensive picture of tools in this area has been done.

In all cases, it is important that there is clear correspondence between the eligibility criteria elements (discussed later) and the review's objective and questions (eg, a scoping review focused on pediatric [participants] assessment tools [concept] in cancer care in Australia [context] must necessarily detail each of these key PCC elements). A benefit of having clear and precise review questions is that it helps authors to think about what specific details they need to collect to meet the objective of their proposed review. This will make it easier to both develop a clear and practical inclusion criterion as well as tailor data extraction tools.

Introduction

The introduction section (or background) of the scoping review protocol is where the bulk of the rationale for conducting the review is presented. Here, the authors should provide a clear explanation of why it is important for a scoping review to be conducted with their objective in question. For example, what issues in clinical practice and research have led to the need for a comprehensive picture of screening and assessment tools for pediatric patients with cancer? It might be that the authors are aware that clinicians have unmet needs around selecting the most appropriate tools, or that there may be little clear and accessible evidence regarding what tools have been validated and how.

As a guide, if the objective is not included as an element of the introduction itself, the introduction should begin with a brief overview to the area under investigation so that the reader has a clear picture of the field in which the work sits. Here, the authors should provide a short review of what is known in the field; a detailed review is not usually required, so citations to key references will normally be acceptable. Once the reader has been provided with a clear overview of the field in question, the authors should then detail the particular problem or issue that has led to the need for the scoping review. The reader will then be able to gain a clear understanding of why the scoping review is required and what needs it fulfills in the field. This early introductory section will also generally provide details regarding the specific inclusion criteria of the scoping review

(described later). This ensures that the reader understands the rationale for the specific inclusion criteria in relation to the field and specific problem or issue under investigation. This section normally echoes the objective and questions, but does not present them verbatim; rather, it explains *why* the objective and questions are such. This section also allows the introduction of new terms and definitions that are specific to the topic researched. It is also an opportunity to introduce the scope of the research question and its boundaries.

Once the topic has been introduced and the need for the review presented, the authors should then explain why a scoping review is justified and appropriate as a method of inquiry. This demands some understanding on the part of the authors regarding what a scoping review can and cannot do, and returns to the broad nature of scoping reviews themselves. The authors might explain that prior to establishing the most effective treatment for oral mucositis in children (which would be addressed by a specific systematic review), it is vital to have a complete picture of all the treatments for oral mucositis (which a scoping review could determine). This kind of detail allows the reader to establish whether the authors have picked “the right tool for the job” (and implies that the authors also know how to use it!).

Finally, the introduction will provide details regarding whether existing scoping reviews or similar systematic reviews have been or are currently being conducted on the topic. Ideally, this is presented along with the date/s and location/s of where searches for existing work has occurred. Partly, this is to demonstrate that the proposed work will not duplicate existing work and partly it provides an opportunity for authors to explain whether their work is entirely new and novel or otherwise fits in with—and indeed extends—current understandings and knowledge in the field. If there are existing similar reviews, then these may have been introduced earlier in the background. If the protocol has been registered, the registration number or publication citation or link can also be included here. We note that the PROSPERO database does not officially register scoping review protocols or reviews,²² but that a number have misleadingly been registered as systematic reviews.^{23,24}

Eligibility criteria

Overall, the eligibility criteria (or inclusion criteria) for scoping reviews are constructed similarly to systematic reviews. These criteria relate to the review's objective and questions and are used to determine what is included/excluded in the review. Those who are familiar with systematic reviews will recognize the PICO framework (participants, intervention, comparator, outcomes). For systematic reviews, each of these are usually precisely defined, as the objective of the review is similarly precise (eg, to establish the effectiveness of Intervention A versus Comparator B among Participant group C in relation to Outcomes D–F). As we know, scoping reviews do not have the same kinds of objectives as systematic reviews, and as such, the elements of their eligibility criteria are different. Overlap and repetition between details provided under each element of the eligibility criteria should be avoided (eg, details about participants should not also be provided under the context).

Participants

This element may also sometimes be called “population.” Important characteristics of the participants included in potentially relevant sources of evidence should be clearly detailed here, including factors such as age, diagnosis, role (eg, nursing staff), and any other qualifying criteria that make a particular participant group appropriate for the objective and questions of the scoping review. Enough detail should be provided so that the authors are able to easily, and ideally unequivocally, distinguish between participants who are eligible for inclusion and those who are not. For this reason, if, for example, a specific diagnosis is required, exactly how that diagnosis was made should be provided (eg, clinical, self-report, histological). Authors may choose to be more inclusive and include all forms of diagnosis and even papers that simply state that participants had the illness in question. It should be clear to the reader how the authors will identify participants of interest and that these participants are clearly relevant in relation to the objective and questions of the review; the exact nature of how participant details are reported is less important.

Justification for the inclusion or exclusion of participants should be explained. Confounding population factors, such as comorbidities or coexisting

conditions (eg, pregnancy), can also be detailed here as exclusion criteria. It is also important to note that scoping reviews may not be focused on participants as such. For example, a scoping review on how the concept of “integrated health care” has been used in policy documents is unlikely to require the details of a particular participant group to be provided as an element of the inclusion criteria.

Concept

The concept of a scoping review is the key issue or topic that the scoping review will explore. This could include (but is not limited to) definitions, methodological approaches, study design, theories, interventions, programs, and conduct decisions. The concept element of a scoping review could also be termed the focus of the proposed review and is usually where details most central to the objective and questions of the review are provided. The core concept examined by the scoping review should be clearly articulated to guide the scope and breadth of the inquiry. This may include details that pertain to elements that would be provided in a standard systematic review, such as the interventions (eg, surgical approaches, pharmacological and/or non-pharmacological therapies), and/or phenomena of interest (eg, the experiences or perspectives of participants), and/or outcomes (eg, quality of life, patient-reported outcomes). For example, screening and assessment tools, which in a traditional systematic review may be conceptualized as interventions (as an approach to measuring outcomes), would be part of the concept of a scoping review designed to map the types, details, and outcome domains of existing tools. It would then be necessary to provide any relevant details pertaining to the tools that may be important for the review, such as whether only particular tools are to be investigated (eg, validated tools only) or whether any or all tools are eligible for inclusion.

Another example are scoping reviews that focus on the ways that research is conducted within a particular field or context. For example, a recent scoping review examined how research has been undertaken in the palliative and end-of-care field in Australia.²⁵ Such a review may provide information in the concept regarding the types of methodologies, theoretical models, methods, and approaches that are to be examined.

Context

The context element of a scoping review's eligibility criteria generally relates to the location and/or field of the concept and/or participants of the review.¹² The context may refer to the physical location of where the sources of evidence (in this case, studies) have been carried out (eg, in primary health care contexts including hospitals, general practices, and specialist health services). Because "context," "setting," and "environment" are often used interchangeably, it is also important to identify other words in the eligibility criteria and literature search that could describe the context (eg, hospitals are also referred to as "tertiary institutes"). Context may also refer to geographic location within particular countries or regions (eg, low- to middle-income countries, or regional and remote areas).¹² The field or area in a thematic sense may be incorporated under context as in "cancer care services," "pediatrics," or "allied health." Context may also be social or cultural (eg, Indigenous, Aboriginal, First Nations, culturally and linguistically diverse communities).¹² The context element may also include considerations of timing (eg, within the context of contemporary health care practice over the previous five years, preoperative contexts, in the context of gender empowerment research over the previous 30 years). In all cases, it is important that the authors provide clear working definitions of key terms, such as "low to middle income," "specialist health services," and/or "Indigenous." In a recent scoping review,²⁶ the authors clearly specified their context (both geographic classification and temporal) in the title and eligibility criteria:

Scoping review to identify and map the health personnel considered skilled birth attendants in low-and-middle income countries from 2000–2015

Types of evidence sources

For the purposes of a scoping review, evidence sources can include any literature (eg, primary studies, systematic reviews, meta-analyses, letters to the editor, guidelines, websites, policy documents). Authors may choose to leave the source of information open to allow for the inclusion of all sources. Otherwise, limits may be put on the types of eligible sources. This may be done with the knowledge that certain sources are most useful and appropriate for a particular topic. For example, a scoping review could be conducted that focuses only on policy

documents and government reports. Authors of scoping reviews are likely to encounter situations where duplicate data are identified as a result of including both primary sources and evidence syntheses that have included the primary source. Here, it is recommended that the authors transparently report in the protocol how they will handle this occurrence during the conduct of their review (eg, primary sources will be excluded if already incorporated into an included evidence synthesis unless the data they contain are not otherwise reported in the evidence synthesis).

Methods

Authors should state at the opening of the methods section what methodology or framework (eg, JBI,¹² Arksey and O'Malley,¹ Levac *et al.*⁸) they will be following. For example, they may include a sentence such as, "The proposed scoping review will be conducted in accordance with the JBI methodology for scoping reviews."

Search strategy

At the protocol stage, authors should begin to plan and develop their intended approach to the search for and selection of potential evidence sources to be included in the final review report. It is suggested that authors recruit the expertise of a research librarian or information scientist in the development of a search strategy and during the search. Details of the collaboration with an information scientist or research librarian should be written in the protocol if this is planned, and a discussion should ideally take place around acknowledging or listing them as an author. The level of detail set out at this stage may range from a simple list of key terms and words based on the inclusion criteria to a more advanced search strategy to be deployed across the entire range of databases to be examined. For publishing purposes, some journals may have requirements regarding the level of detail expected at the protocol stage. For example, *JBI Evidence Synthesis* requires one full search strategy to be included as an appendix to the submitted protocol.

The process of searching for sources for inclusion relies on an appropriate range of relevant keywords and terms. Deciding which keywords and terms to use may be challenging because of variations in terminology used between the various sources. This challenge can be compounded when adapting search

strategies for different databases with varied taxonomies and indexing terms. As in systematic reviews, only those databases that contain literature relevant to the review should be searched; for example, a scoping review that seeks to provide information for low-income countries should include the LILACS database.²⁷ Ensuring clarity in the articulation of the inclusion criteria is vital for the creation of a robust search strategy. Authors may choose to develop a concept map based on the key terms articulated in the inclusion criteria to establish a comprehensive list of synonyms and search terms.

Because of the necessary breadth of searches required to comprehensively map a body of literature, defining the boundaries of the scope of the search, particularly when searching for gray literature sources, can be a challenge in the development of searches for scoping reviews. Authors should provide a detailed list of databases, search platforms, relevant institutional websites, and any other literature sources that are likely to contain literature relevant to the scoping review. The choice to search more or fewer sources may be guided by decisions around balancing the breadth of the search with the time and resources allotted to the project. On one hand, the exclusion of potentially relevant literature sources may impose limitations on the comprehensiveness of the review, but these can be transparently acknowledged and rationalized. On the other hand, when inclusion criteria and the search are not sufficiently specific, authors can encounter difficulties in consistently including or excluding the right studies from a large pool of literature. This can make the process of study screening and selection time consuming and potentially unachievable.

Beyond detailing some level of the planned search terms and literature sources to be examined, this section of the protocol should also clearly identify, with justification, the earliest date that the search is planned to include. This date may be the point when the first mention of key concepts, interventions, or issues to be investigated by the scoping review were identified in the literature. Some scoping reviews may focus on mapping evidence within a finite and defined period (eg, the most up-to-date and clinically relevant evidence only). In any case, there should be transparent and clear rationalization for the decision to exclude potential sources of evidence from the review based solely on the time of publication.

In this section, the authors should also provide some detail regarding the approach to the search; for instance, whether an initial search is to be undertaken across a limited range of databases in order to identify potentially relevant keywords and terms for developing a final search strategy across all databases, whether the reference lists of relevant sources of evidence will be examined for additional references, and whether key authors will be contacted with requests to provide potentially relevant sources. If the authors plan to use a particular software for documenting and managing the results of the search, this should be identified.

Study/source of evidence selection

As with systematic reviews, the selection of sources of evidence located by the search is based solely on whether they meet the protocol's inclusion criteria. Authors should provide some detail in the protocol regarding how this will be established. This may involve screening identified sources of evidence first at the level of the title and/or abstract followed by a full-text examination of sources that appear to be relevant. The authors should state how many members of the review team will carry out the screening and selection of evidence sources and whether this process will be cross-checked by other authors (eg, by two or more authors screening/selecting all or a subset of potentially relevant sources to determine agreement). The authors should also state whether they will pilot the screening process on a subset of potentially relevant sources, how many or what percentage of sources will be included in the pilot, and how many authors will participate. This piloting exercise can be useful for large review teams and help to ensure consistency between authors. A statement of how disagreements will be managed should also be included in this section. Typically, disagreements are managed through either discussion between the two data screeners or by including a third party to decide. Any software used to support this process (ie, JBI SUMARI) should be mentioned.²⁸

The protocol should provide some information regarding how excluded sources will be reported in the final review (eg, whether they will be included in an appendix with reasons for exclusion reported, and whether this list will include all sources excluded or only those excluded at the full-text level). The authors may choose to detail the planned use of an adapted PRISMA 2020 flow diagram²⁹ or the

adapted chart in the PRISMA-ScR⁵ to graphically depict the movement of sources through the search process to eventual inclusion.⁵

A final point on writing the planned approach for searching and selection at the protocol stage of a scoping review is that because of the broad nature of this type of review, it is likely that searching, screening, and selection may reveal new potentially relevant terms, concepts, and even locations of evidence. As such, the authors may choose to state that the search may be modified and expanded during the review process to account for these eventualities.

Data extraction

The purpose of extracting the evidence is to detail the results of the scoping review in a clear and easy-to-understand fashion. Beyond extracting basic descriptive data about the included sources of evidence (eg, authors, title, year published), the data to be extracted should be consistent with the review question/s and the inclusion criteria. A draft table detailing the data to be extracted should be included in the protocol with fields for data items that align with the review question/s, participants, concept, and context described previously (see Table 1). It should be apparent to the reader that the authors have carefully considered what data are required to be extracted from the included evidence sources to answer the review's question/s. The authors may state that the draft extraction tool will be piloted on a subset of sources to be included in the review to test its feasibility for the review. This can then be refined further to address the research question for the scoping review as required. The authors should state if the final completed tools will be presented in the scoping review report. The authors should also state how many authors will be involved in data

extraction, whether extraction will occur independently, whether cross-checking of extracted data will occur, and how disagreements will be settled between authors regarding dissimilarities in terms of data extraction.

Unlike traditional systematic reviews, scoping reviews do not typically include a step for the assessment of the methodological quality or risk of bias of sources of evidence. In the rare case where assessment of methodological quality is included in a scoping review, it should be clearly justified to meet the particular review's objective/s or review question/s. If the authors choose to include this optional element, this should be detailed in the section that reports on the extraction of data along with the proposed approach to assessment and what tools may be used. Because scoping reviews seek to comprehensively map a body of literature, the results of methodological quality assessment should not be used to determine inclusion/exclusion from the review.

Data analysis and presentation

This section should contain details about how the review authors intend to analyze and present an overview and description of all the included sources as well as the data extracted from them. Scoping reviews do not undertake data synthesis such as statistical meta-analysis, which should only occur following assessment of the methodological quality, heterogeneity, or risk of bias of included studies. Analysis of data in scoping reviews is normally descriptive, with basic frequency analysis and percentages being the most common approach. There should be a description of how the results will be presented (eg, with maps, diagrams, tables). Scoping reviews aim to identify the existing knowledge, thus

Table 2: Example of a basic draft extraction tool

Source of evidence (citation)	Year	Country	Participants	Cancer			Treatment/s	Screening tool/s (+ validated Y/N)	Assessment tool/s (+ validated Y/N)
			Sex	Age	Type	Stage			

it is important to present conceptual categories such as intervention type, study population (and sample size, appropriate), duration of intervention, aims, methodology adopted, key findings (evidence established), and gaps in the research. Due to the broad and often iterative nature of scoping reviews, decisions around how the data are ultimately presented may change and evolve. At protocol stage, it is only necessary to provide some indication that the authors have considered how to best present their anticipated results in a manner that will rationally correspond with the objective/s and review question/s.

Deviations from the protocol

A detailed a priori protocol should be followed closely. However, scoping reviews, especially within the search strategy, data extraction, and data presentation stage, tend to be an iterative process. Because of this, it is common for authors to deviate from the protocol. If this occurs, authors should clearly mention these deviations within their review manuscript to ensure the process of the conduct of the scoping review remains transparent.

Conclusion

This paper has introduced the purpose and importance of developing a detailed a priori scoping review protocol before the conduct of a full review. The recommended elements or sections that should ideally be presented in a scoping review protocol have been discussed in detail with examples and descriptions provided to illustrate how to clearly articulate and explain the planned purpose, aims, inclusion criteria, and process of carrying out a scoping review. A basic example template is available in the JBI Manual for Evidence Synthesis, and Appendix I contains a fillable checklist with the items listed for a scoping review protocol found in Table 1.¹² Some examples of recent scoping review protocols that have been reported in line with these items are cited here.³⁰⁻³² As we know, knowledge users rely on clear and transparent reporting to understand and use the results of published work to drive evidence-based improvements within health care and beyond. It is hoped that readers will be able to use this paper to assist with planning future scoping reviews and to carry them out with a high degree of rigor and transparency.

Acknowledgments

Cassia Baldini Soares from the University of Sao Paulo, School of Nursing, for her feedback on an early draft of this paper.

Funding

No funding was received by any author specifically relevant to this manuscript. ACT is funded by a Tier 2 Canada Research Chair in Knowledge Synthesis. ZM is supported by an NHMRC Investigator Grant ID:1195676.

Declarations

The authors of this manuscript are the authors of the JBI Methodology for the Conduct and Reporting of Scoping Reviews. MDJP and CG are also coauthors and contributors to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses: extension for Scoping Reviews (PRISMA-ScR).

Author contributions

All authors contributed to the authorship of the manuscript and are listed in order of contribution.

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Appendix I: Recommended items to address in a scoping review protocol*

Section and topic	Item No	Checklist item	Page#
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a scoping review	
Update	1b	If the protocol is for an update of a previous scoping review, identify as such	
Registration	2	If registered, provide the name of the registry (such as JBI) and registration number	
Authors:			
Contact	3a	Provide name, institutional affiliation, email address of all protocol authors; provide physical mailing address of corresponding author	
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	
Support:			
Sources	5a	Indicate sources of financial or other support for the review	
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known (Note: Consider providing a rationale for the choice of conducting a scoping review as compared to other evidence synthesis approaches)	
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to the inclusion/ exclusion criteria	
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, timeframe) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other gray literature sources) with planned dates of coverage	
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion)	
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any preplanned data assumptions and simplifications (Note: Scoping reviews may not use PICO and instead may use JBI's Population, Concept, and Context [PCC] or another approach to reporting eligibility criteria)	
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale (Note: Scoping reviews may not extract outcome data, so this can refer to whichever data items are extracted)	

(Continued)			
Section and topic	Item No	Checklist item	Page#
Risk of bias in individual studies	14	If this is to occur, describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis (Note: Scoping reviews typically do not include risk of bias assessment, but this information should be described if it will occur)	
Data synthesis	15a	Describe criteria under which study data will be presented (Note: Scoping reviews do not typically include quantitative synthesis of study data, but should still describe in advance how extracted data are anticipated to be presented in the resulting review)	
	15b	Describe the planned approach to how extracted data will be presented (such as figures, tables, evidence gaps maps)	
	15c	Describe any proposed additional analyses (such as thematic analyses) (Note: The JBI methodological guidance does not recommend undertaking thematic analysis as this synthesis of data should ideally occur following methodological appraisal of the included sources)	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) (Note: Scoping reviews typically do not include assessment of meta-bias(es), but this information should be described if it will occur)	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	

GRADE, Grading of Recommendations, Assessment, Development and Evaluation; PICO, participants, intervention, comparator, outcomes; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, et al.; PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015;349:g7647. CC BY 4.0.

"—This checklist is based on the PRISMA for systematic review protocols (PRISMA-P) checklist.ⁱ It is strongly recommended that this checklist be read in conjunction with the best practice guidance and reporting items for the development of scoping review protocols for important clarification on the items, ⁱⁱ and the JBI updated methodological guidance for the conduct of scoping reviews.ⁱⁱⁱ Amendments to a scoping review protocol should be tracked and dated.

ⁱMoher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. *Syst Rev*. 2015;4(1):1.

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